UNIVERSITI TEKNOLOGI MARA

TEST (SET II)

NAME	:	
STUDENT NO.	:	
GROUP	:	
LECTURER	:	
DATE	:	
COURSE AND COURSE CODE	:	INTRODUCTION TO PROBABILITY AND STATISTICS (STA116)
TIME	:	2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of three (5) questions.
2. Answer ALL questions.
3. Do not refer any materials and make sure the answer script is completed before submit.

For Lecturer's Use Only:

QUESTION	MARKS
Q1 (10M)	
Q2 (10M)	
Q3 (10M)	
Q4 (10M)	
Q5 (10M)	
TOTAL (50)	

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QUESTION 1

(a) Determine each of the following variables' level of measurement (nominal, ordinal, interval, ratio).

i. Brand of shoes collected by Ali.

(1 mark)

ii. Level of difficulty in a mobile game application.

(1 mark)

iii. Salary range of workers in UiTM Sabah based on occupation grades.

(1 mark)

iv. CDCS110 students' favourite song genres.

(1 mark)

v. Number of pages in a statistical textbook.

(1 mark)

(b) Starting 1st July 2025, sales and service tax is re-introduced for implementation in all economic sectors and markets. A researcher wants to conduct a study on customers' satisfaction towards the services and delicacies provided in cafes around Kota Kinabalu, Sabah. A total of 10 cafes will be selected out of 350 cafes. The variables aim to be collected include the number of reviews done on each café, the ratings given for café, and type of menus served.

i. Describe the population in this study.

(1 mark)

ii. State the most appropriate sampling technique that should be used.

(1 mark)

iii. Describe three types of variables will be chosen for the study and their level of measurement.

(3 marks)

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QUESTION 2

(a) The following are the number of UK tourists coming to Kota Kinabalu in the first 8 months of year 2024.

86, 92, 88, 95, 90, 93, 94, 92

Compute the following for the data:

i. Mean and median (2 marks)

ii. Variance and standard deviation (3 marks)

(b) The following table shows the number of sales at a clothes store per day for jeans and t-shirts:

Day	Jeans	T-shirt
Monday	5	12
Tuesday	8	7
Wednesday	3	9
Thursday	6	15
Friday	17	18
Saturday	20	13

Draw a stacked bar chart to represent the above data. (5 marks)

QUESTION 3

(a) A musician wants to create a unique 3-note melody using the notes C, D, F, G, and B. No note can be repeated in the melody. How many different 3-note melodies can be formed?

(2 marks)

(b) The word STATISTICS is printed on ten individual tiles (one letter per tile). How many arrangements can be possibly made from this word?

(3 marks)

(c) In a training session, there are 12 participants: 7 are from Department A and 5 are from Department B. If the organiser randomly pick 4 participants to receive a free gift:

i. What is the probability that no more than one participant is from Department B?

(3 marks)

ii. What is the probability that all selected participants are from Department A?

(2 marks)

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QUESTION 4

(a) At a university, students submit their assignments through three different platforms: **e-Learn, Google Classroom,** and **Email**. Records show that 50% of students use e-Learn, 30% use Google Classroom, and the remaining 20% submit via Email. However, submission issues (such as files not uploading properly) occur with different probabilities for each platform: 4% for e-Learn, 7% for Google Classroom, and 3% for Email.

i. Draw a tree diagram to represent this situation.

(2 marks)

ii. If a submission issue is detected, what is the probability that it came from Google Classroom.

(3 marks)

(b) A chocolate factory has three machines: **Machine R**, **Machine S**, and **Machine T**.

- Machine R produces 1500 chocolates/day
- Machine S produces 2000 chocolates/day
- Machine T produces 2500 chocolates/day

The probability of a chocolate being misshaped is:

- 1.5% from Machine R
- 2% from Machine S
- 0.5% from Machine T

i. Find the probability that a randomly selected chocolate is misshaped.

(2 marks)

ii. If a chocolate is found not misshaped, what is the probability it was produced by Machine S?

(3 marks)

QUESTION 5

(a) In a pencil case, there are **4 mechanical pencils, 6 gel pens,** and **10 ballpoint pens**. Two writing tools are picked one after another without replacement.

i. What is the probability that both items selected are ballpoint pens?

(2 marks)

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ii. What is the probability that the first is a gel pen and the second is a mechanical pencil?

(3 marks)

(b) Students attending a **Public Speaking Bootcamp** or a **Writing Bootcamp** show the following success rates:

- 65% of students attend the Public Speaking Bootcamp
- 35% attend the Writing Bootcamp
- 80% of Public Speaking attendees complete the course
- 90% of Writing attendees complete the course

i. What is the probability that a randomly selected student completes the course?

(2 marks)

ii. Given that the student completed the course, what is the probability they had attended the Writing Bootcamp?

(3 marks)

END OF QUESTION PAPER

FORMULA LIST

Sample measurements

1. Mean, $\bar{x} = \frac{\sum x}{n}$ or $\frac{\sum fx}{n}$

2. Median, $\tilde{x} = L_m + \left(\frac{\frac{n}{2} - \sum f_{m-1}}{f_m} \right) \cdot C$

3. Mode, $\hat{x} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \cdot C$

4. Standard Deviation, $s = \sqrt{\frac{1}{n-1}\left[\sum (x - \bar{x})^2 \right]}$ or $\sqrt{\frac{1}{n-1}\left[\sum x^2 - \frac{(\sum x)^2}{n} \right]}$

$\sqrt{\frac{1}{n-1}\left[\sum f(x - \bar{x})^2 \right]}$ or $\sqrt{\frac{1}{n-1}\left[\sum fx^2 - \frac{(\sum fx)^2}{n} \right]}$

5. Coefficient of variation, $cv = \frac{s}{\bar{x}} \times 100$

6. Pearson's measure of skewness = $\frac{\text{mean} - \text{mode}}{\text{standard deviation}}$ or $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$

where

n	=	$\sum f$
L_m	=	lower median class boundary
L_{mo}	=	lower modal class boundary
$\sum f_{m-1}$	=	cumulative frequencies for the classes before the median class
f_m	=	median class frequency
Δ_1	=	(modal class frequency) - (frequency for the class before the modal class)
Δ_2	=	(modal class frequency) - (frequency for the class after the modal class)
C	=	class size

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Probability

If A and B are two events,

1. Additive rule 1 (not mutually exclusive events)
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2. Additive rule 2 (mutually exclusive events)
$$P(A \cup B) = P(A) + P(B)$$

3. Multiplication rule 1 (independent events)
$$P(A \cap B) = P(A) \cdot P(B)$$

4. Multiplication rule 2 (dependent events)
$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

5. Conditional probability
$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

6. Bayes' Theorem
$$P(B_i \mid A) = \frac{P(B_i \cap A)}{P(A)} \quad \text{for } i = 1,2,...,k$$
where $P(A) = P(B_1) \cdot P(A \mid B_1) + P(B_2) \cdot P(A \mid B_2) + ... + P(B_k) \cdot P(A \mid B_k)$

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